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Setup and configure LAN using a Switch and ethernet cables

Aim :-

Setup and configure a LAN (Local Area Network) using switch and ethernet cables in lab.

→ LAN :-

A local area network (LAN) refers to a network that connects devices within limited area.

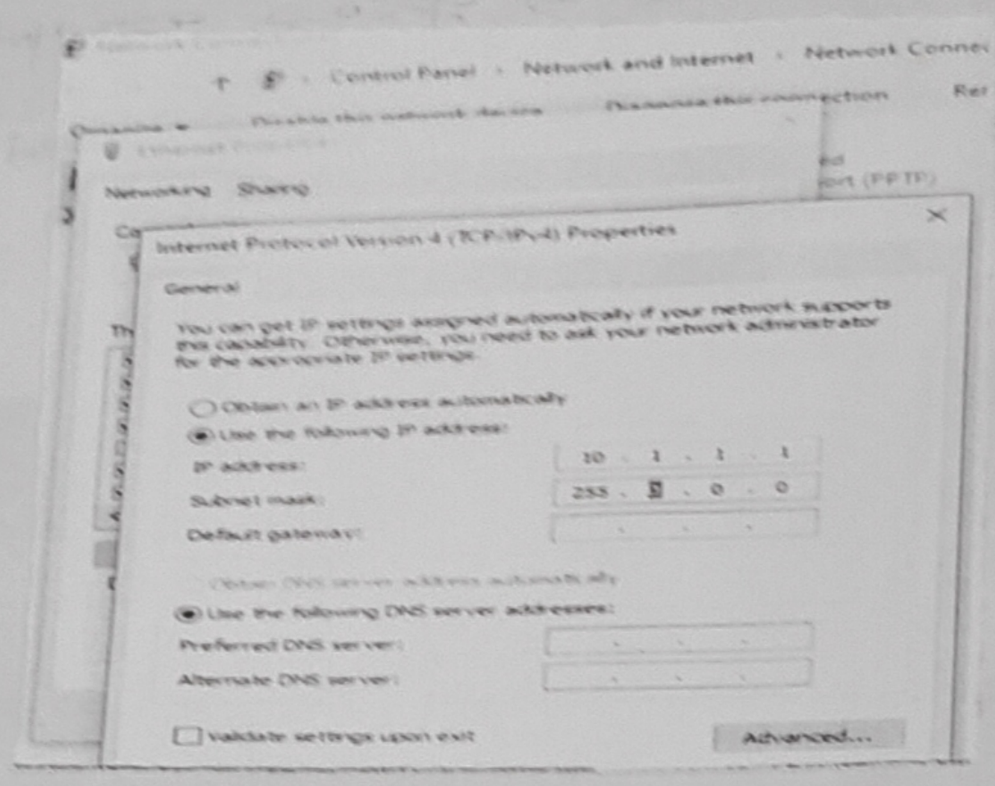
→ How to set up a LAN :-

- Plan and design appropriate network topology taking into account network.
- You can take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes.
- Connect your computers to network switch via ethernet cable, which is as simple as plugging one end of ethernet cable into network switch.
- Assign IP address to your PCs

  log on to computer

  click network and internet connections

  Right click local area connection → select internet protocol


Control Panel › Network and Internet › Network Connections
Networking Sharing
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 10 . 1 . 1 . 1
Subnet mask: 255 . 0 . 0 . 0
Default gateway:
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...

- Configure a network switch :
  Connect computer to switch
  Log in to web interface
  Configure Basic settings
  Assign IP address

- check the connectivity Between switch and other machine By using ping command in the command prompt of device.

- select a folder → go to properties → click sharing tab → share it with everyone on the same LAN.

- Try to access the shared folder from others computers of network.

Result :

Hence setup and configure LAN using a switch and ethernet cables has Been executed successfully.